

BT611P

Assignment 2, Fall 24

Total Marks: 10

Due date: 6th December 202

Question 1:

Digital Diagram of Preparation of Serial Dilutions for Isolation and CFU calculation of bacterial cells.

Objective:

Create a detailed **digital-only diagram** illustrating the **serial dilution process for isolation of bacteria**. The diagram should cover the key steps involved in isolation processes CFU calculation.

Assignment Scope:

This assignment will help you understand the significance of serial dilution process and will enable you to see importance of CFU calculation in estimating bacterial load in any sample. It will also enhance your digital drawing skills by requiring a clear, informative visual representation of each stage without the use of text or external images.

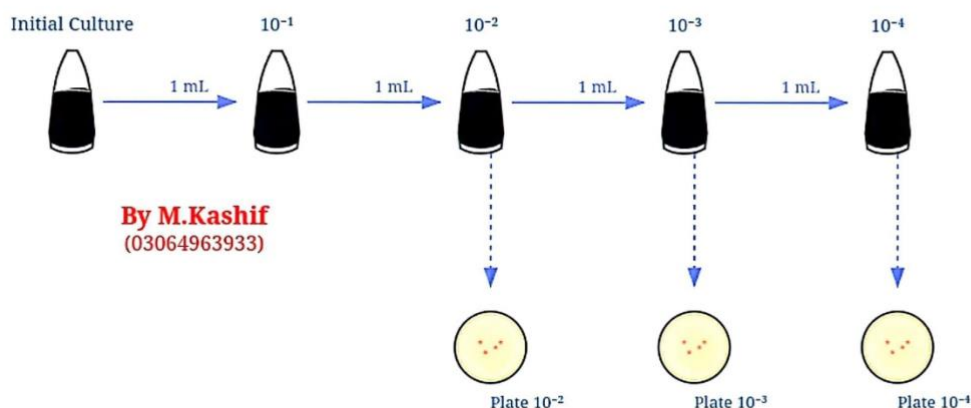
Guidelines and Requirements

1. **Software and Tools:** Use digital drawing software or tools of your choice (e.g., Adobe Illustrator, Microsoft PowerPoint, Canva, Paint or any other tools you like). Ensure that your final product is a clean, high-resolution digital illustration.
2. **Format:** Submit your final diagram and ensure the file resolution is clear and suitable for printing or display.
3. **No Text:** Submissions without any digital diagram will not be considered and will be marked zero.
4. **Original Digital Artwork Only:** All components of the diagram must be **created by you** in a digital format. Hand-drawn, AI-generated, or any internet-sourced images will not be accepted.



5. **Visual Clarity:** Use distinct colors, shapes, symbols, and arrows to make each stage of microbial activity clear. Consider incorporating flow indicators to demonstrate progression in spoilage, fermentation, and product transformations.
6. **Creativity and Accuracy:** You are encouraged to apply creative digital design techniques to accurately represent scientific processes. Ensure each microbial activity is correctly illustrated and relevant to dairy fermentation or spoilage.
7. **Assessment Criteria:**
 - **Scientific Accuracy:** The diagram should accurately represent serial dilution processes.
 - **Clarity and Detail:** Each stage of microbial activity should be distinct and visually clear.
 - **Creativity:** Effective use of visual elements to convey complex processes without text.
 - **Originality:** All diagram components are original digital drawings created by you.

Serial Dilution Process for CFU Calculation



CFU Calculations

Formula: $\text{CFU/mL} = (\text{Number of colonies} \times \text{Dilution factor}) \div \text{Volume plated}$

Example Calculation:

• If 45 colonies are counted on the 10^{-4} dilution plate:

$$\text{CFU/mL} = (45 \text{ colonies} \times 10^4) \div 0.1 \text{ mL} = 4.5 \times 10^6 \text{ CFU/mL}$$

Key Steps:

1. Mix thoroughly between transfers
2. Plate 0.1 mL on each plate
3. Incubate at 37°C for 24-48 hrs
4. Count plates with 30-300 colonies

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